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Intermolecular rovibrational states of the $\text{H}_2\text{O}-\text{CO}_2$ and $\text{D}_2\text{O}-\text{CO}_2$ van der Waals complexes **FREE**

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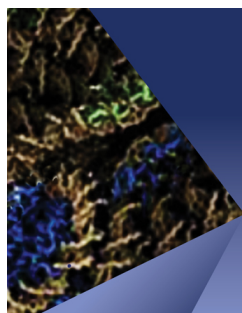


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ABSTRACT

We present quantum five-dimensional bound-state calculations of the fully coupled intermolecular rovibrational states of $\text{H}_2\text{O}-\text{CO}_2$ and $\text{D}_2\text{O}-\text{CO}_2$ van der Waals (vdW) complexes in the rigid-monomer approximation for the total angular momentum J values of 0, 1, and 2. A rigid-monomer version of the recent *ab initio* full-dimensional (12D) potential energy surface of $\text{H}_2\text{O}-\text{CO}_2$ [Q. Wang and J. M. Bowman, J. Chem. Phys. 147, 161714 (2017)] is employed. This treatment provides for the first time a rigorous and comprehensive description of the intermolecular rovibrational level structure of the two isotopologues that includes the internal-rotation tunneling splittings and their considerable sensitivity to rotational and intermolecular vibrational excitations, as well as the rotational constants of the two vdW complexes. Two approaches are used in the calculations, which differ in the definition of the dimer-fixed (DF) frame and the coordinates associated with them. We demonstrate that with the approach introduced in this work, where the DF frame is fixed to the CO_2 moiety, highly accurate results are obtained using significantly smaller basis sets in comparison to those for the alternative approach. In addition, the resulting wavefunctions tend to lend themselves better to physical interpretation and assignment. The $\text{H}_2\text{O}-\text{CO}_2$ ground-state internal-rotation tunneling splittings, the rotational transition frequencies, and the rotational constants of both vdW complexes are in excellent agreement with the experimental results. The calculated intermolecular vibrational fundamentals agree well with the scant terahertz spectroscopy data for these complexes in cryogenic neon matrices.

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I. INTRODUCTION

H_2O and CO_2 are abundant in the atmosphere of our planet, the latter, a major greenhouse gas, in ever increasing concentrations. Quantitative characterization of the interaction between these two molecules is important in several contexts of fundamental significance. One of them is achieving molecular-level understanding of the dissolving of CO_2 in water leading to the formation of the carbonic acid (H_2CO_3), which governs the exchange of CO_2 between Earth's atmosphere and the oceans (and is also important for the CO_2 transport in the blood). The formation of the $\text{H}_2\text{O}-\text{CO}_2$ complex can be viewed as the initial step in these processes. The $\text{H}_2\text{O}-\text{CO}_2$ van der Waals (vdW) complex is involved also in the photosynthetic conversion of CO_2 and H_2O to organic molecules, which sustains the life on the Earth. Last but not least, this

fluxional complex exhibits a most interesting vibration-rotation-tunneling dynamics, which has attracted considerable attention from experimentalists and theorists alike.

The first microwave spectra of the $\text{H}_2\text{O}-\text{CO}_2$ complex, and those of the two isotopologues $\text{D}_2\text{O}-\text{CO}_2$ and $\text{HDO}-\text{CO}_2$, were measured by Peterson and Klemperer¹ using molecular-beam-electronic-resonance spectroscopy. From them, it was deduced that the complex has a planar T-shaped equilibrium structure with C_{2v} geometry, where the oxygen atom of water is weakly bound to the carbon atom of CO_2 and the two hydrogen atoms point away from CO_2 . It was also concluded that the two monomers rotate internally relative to one another around the C $\cdots$ O vdW bond (which coincides with the a principal axis of the complex), with the estimated internal-rotation barrier of $315(70) \text{ cm}^{-1}$. However, these experiments could not directly probe the tunneling splittings caused

by this internal rotation, owing to the electric dipole selection rule associated with the pure rotational spectrum. Direct measurement of the internal-rotation tunneling splitting was accomplished by Block *et al.*² from a rotationally resolved infrared (IR) spectrum of $\text{H}_2\text{O}-\text{CO}_2$ in a molecular beam, giving the value of $0.1356(13) \text{ cm}^{-1}$ and the estimate of $315(70) \text{ cm}^{-1}$ for the barrier to internal rotation. Besides the estimates of the internal-rotation barrier height and the tunneling splitting, the microwave spectroscopy of $\text{H}_2\text{O}-\text{CO}_2$ and its H/D isotopologues^{1,3} yielded a large number of rotational transition frequencies, as well as the rotational constants and other structural information, for the ground vibrational states of the complexes. The $\text{H}_2\text{O}-\text{CO}_2$ and $\text{D}_2\text{O}-\text{CO}_2$ complexes prepared by supersonic expansion have also been studied using high-resolution IR spectroscopy in the regions of the OH stretch fundamentals² and overtones^{4,5} and the ν_2 bend region of the water monomer.⁶ Finally, the $\text{H}_2\text{O}-\text{CO}_2$ complex (and its H/D isotopologues) embedded in the cryogenic Ne matrices was investigated in the far-IR^{7,8} and mid-IR⁸ regions. In particular, Andersen *et al.*⁷ observed three intermolecular vibrations of the complex, all with energies above 100 cm^{-1} .

The stationary points, energetics, and harmonic and anharmonic vibrational frequencies of the $\text{H}_2\text{O}-\text{CO}_2$ complex have been the subject of numerous *ab initio* studies (see, for example, Ref. 9). However, this complex is weakly bound and fluxional, exhibiting coupled large-amplitude motion (LAM) intermolecular vibrations and, as evident from experiments, small but readily observable tunneling splittings indicating wavefunction delocalization. These and other features pose a serious challenge to theory and cannot be described accurately by considering only the equilibrium geometry of the complex. What is required for this purpose is an *ab initio*-based potential energy surface (PES) of $\text{H}_2\text{O}-\text{CO}_2$, either twelve-dimensional (12D) for flexible monomers or in 5D for the monomers treated as rigid (since CO_2 is linear, it would be 6D if both molecules were nonlinear).

Two PESs of this kind were reported in recent years. One of them is the rigid-monomer 5D intermolecular PES of the $\text{H}_2\text{O}-\text{CO}_2$ complex by Makarewicz¹⁰ based on large-scale *ab initio* calculations. It was used to characterize the minimum-energy structure of the complex and identify other stationary points, but no quantum bound-state calculations have been performed on this PES. The second more recent *ab initio* PES of $\text{H}_2\text{O}-\text{CO}_2$ by Wang and Bowman¹¹ is full-dimensional (12D). It was constructed as a permutationally invariant polynomial (PIP) fit to tens of thousands of two-body interaction energies, combined with spectroscopically accurate potentials for non-interacting flexible monomers. The energies were obtained at the coupled-cluster singles doubles perturbative triples [CCSD(T)]-F12b/aug-cc-pVTZ level of theory. A detailed description of this 12D PES can be found in Ref. 11. The PES was employed in the vibrational self-consistent field (VSCF) and virtual state configuration interaction (VCI) calculations of the $\text{H}_2\text{O}-\text{CO}_2$ complex.¹¹ However, while the complex has 12 vibrational modes, these VSCF/VCI calculations were restricted to the seven intramolecular modes, assuming a four-mode representation of the potential. The computed intramolecular fundamentals were found to be within a few wave numbers of the available experimental results. For the intermolecular vibrations of the complex, only harmonic frequencies for the global minimum were reported. They are not expected to be very reliable, given that the intermolecular

vibrations are anharmonic and coupled and have LAM character. The same *ab initio* PIP PES of $\text{H}_2\text{O}-\text{CO}_2$ was also used to develop a potential for CO_2 inside the small cage of the structure II clathrate hydrate, utilized in approximate calculations of the vibrations of the caged CO_2 .¹¹

To date, there have been no rigorous quantum calculations of the intermolecular rovibrational states of the $\text{H}_2\text{O}-\text{CO}_2$ complex. This provides the motivation for the study reported in this paper. In recent years, we have performed full-dimensional and fully coupled quantum calculations of the intra- and intermolecular (ro)vibrational states of several noncovalently bound triatom-diatom complexes, assuming flexible monomers: $\text{H}_2\text{O}/\text{D}_2\text{O}-\text{CO}$,¹² $\text{HDO}-\text{CO}$,¹³ $\text{H}_2\text{O}-\text{HCl}$,¹⁴ and several H/D isotopologues.^{15,16} What we present in this paper are the rigid-monomer quantum 5D calculations of the fully coupled $J = 0, 1, 2$ intermolecular rovibrational states of the $\text{H}_2\text{O}-\text{CO}_2$ and $\text{D}_2\text{O}-\text{CO}_2$ complexes. The rigid-monomer version of the aforementioned 12D PES by Wang and Bowman¹¹ is employed. We resorted to the rigid-monomer approximation for two reasons. First, $\text{H}_2\text{O}/\text{D}_2\text{O}-\text{CO}_2$ is a 12D system, making full-dimensional quantum calculations much more demanding than such calculations in 9D for the triatom-diatom complexes above. Keeping the monomers rigid reduces the dimensionality of the problem to 5D, which is far more manageable. Just as important, the assumption of rigid monomers is not expected to significantly affect the accuracy of the calculated rovibrational eigenstates of the complex and other quantities related to them, as explained below.

In vdW complexes, such as $\text{H}_2\text{O}/\text{D}_2\text{O}-\text{CO}_2$, the coupling between the intermolecular vibrations and the intramolecular vibrations of the monomers is typically weak, especially when the monomers are in their ground intramolecular vibrational states. This is borne out by the results of our recent calculations of two noncovalently bound molecular complexes. In the case of $\text{H}_2\text{O}/\text{D}_2\text{O}-\text{CO}$,¹² with the (9D) binding energy $D_0 = 354.45/407.08 \text{ cm}^{-1}$, for the ground vibrational states of the monomers, the low-lying 9D (flexible-monomer) and 5D (rigid-monomer) intermolecular level energies differ by up to 1 cm^{-1} . For the $\text{HCl}-\text{H}_2\text{O}$ dimer,¹⁴ which is considerably more strongly bound than $\text{H}_2\text{O}-\text{CO}$, having (9D) $D_0 = 1334.63 \text{ cm}^{-1}$, the differences between the flexible-monomer and rigid-monomer intermolecular level energies are on the order of $1-3 \text{ cm}^{-1}$, while for a few, this difference is $4-6 \text{ cm}^{-1}$. The $\text{H}_2\text{O}-\text{CO}_2$ complex, in terms of its binding energy $D_0 = 768.64 \text{ cm}^{-1}$ (from the 5D rigid-monomer calculations in this work), falls between $\text{H}_2\text{O}-\text{CO}$ and $\text{HCl}-\text{H}_2\text{O}$ and is closer to the former. Therefore, we expect the low-lying energies of intermolecular rovibrational states of $\text{H}_2\text{O}-\text{CO}_2$ from the 5D rigid-monomer calculations to be within $1-2 \text{ cm}^{-1}$ of those that would be computed in 12D for flexible monomers (in their ground states).

The rigid-monomer treatment is also expected to yield accurate internal-rotation tunneling splittings. This is based, in part, on the success of the rigid-monomer (4D) quantum treatment of hydrogen-bonded dimers, such as $(\text{HF})_2$, which gave tunneling splittings very close to those from flexible-monomer (6D) calculations.¹⁷ In addition, the tunneling splittings from the more recent rigid-monomer quantum calculations of the water dimer on accurate *ab initio* PESs^{18,19} agree extremely well with the corresponding experimental results.

The results presented here are obtained by two different approaches, both fully coupled. In one of them, introduced by Brocks *et al.*,²⁰ the dimer-fixed (DF) coordinate frame is defined such that its z axis is parallel to the vector pointing from the center of mass (c.m.) of one of the monomers to that of the other. All our previous quantum calculations of the (ro)vibrational states of noncovalently bound triatom–diatom complexes^{12–16} have utilized this approach. In the second approach, we derive and implement the new rovibrational Hamiltonian based on the DF frame embedded in the CO₂ moiety, with its z axis parallel to the internuclear axis of CO₂. We demonstrate that with this novel method, the results equivalent to those from the “Brocks-type” approach²⁰ can be obtained using significantly smaller basis sets. Another important advantage of this approach is that, in general, its results can be interpreted more readily than those obtained by the Brocks-type approach, facilitating the interpretation and assignment of the computed rovibrational states. Therefore, we believe that it should be the method of choice for heavy-light weakly bound molecular dimers.

The rigid-monomer quantum 5D calculations reported in this work provide a rigorous and comprehensive description of the intermolecular rovibrational structures of the H₂O–CO₂ and D₂O–CO₂ complexes that has been lacking until now. Besides the rovibrational eigenstates, this includes their internal rotation tunneling splittings, the $J = 0$ –2 rotational tunneling states of the two complexes in their ground vibrational states, and the rotational constants of the two complexes. A comparison is made with the available experimental data in the literature.

This paper is organized as follows: The computational methodology is described in Sec. II. In Sec. III, we present and discuss the results. Section IV contains the conclusions.

II. COMPUTATIONAL METHODOLOGY

A. General

The computational results we present herein all relate to the intermolecular rovibrational states of water–CO₂ in the rigid monomer approximation. As outlined in the Introduction, two different approaches are employed to obtain those results. In the first, the “Brocks-type” approach, we use the well-known Hamiltonian and associated coordinates from the paper by Brocks *et al.*²⁰ That method arises from the definition a dimer-fixed (DF) coordinate frame in terms of the vector that points from the c.m. of one of the monomer moieties to that of the other.

We also find it advantageous to derive and make use of a new Hamiltonian based on a DF frame fixed to the CO₂ moiety. This latter approach allows for the use of considerably smaller basis sets than those required for Brocks-type calculations. Moreover, the results obtained by this method tend to be more transparent to physical interpretation than those produced by the Brocks-type.

In both methods, we require as input rigid-monomer geometries and inertial parameters. We use the following: for linear and symmetric ¹²C¹⁶O₂, we take the CO bond distance to be 1.162 26 Å, the rotational constant $B = 0.390\,22\text{ cm}^{-1}$,²¹ and the mass to be 43.9892 amu. For water (both H₂¹⁶O and D₂¹⁶O), we take the bond distances to be 0.9575 Å and the bond angle to be 105.2°. These geometrical parameters are close to the equilibrium values of the water monomers.²² They are also consistent with the experimentally determined ground-state rotational constants of the H₂O and

D₂O isotopologues, which we use as input in constructing the kinetic-energy portion of our Hamiltonians. For the rotational constants of H₂O, we use²³ $B_{x_A} = 27.8761\text{ cm}^{-1}$, $B_{z_A} = 14.5074\text{ cm}^{-1}$, and $B_{y_A} = 9.2877\text{ cm}^{-1}$, and for D₂O, we use²⁴ $B_{x_A} = 15.4204\text{ cm}^{-1}$, $B_{z_A} = 7.271\,04\text{ cm}^{-1}$, and $B_{y_A} = 4.846\,81\text{ cm}^{-1}$, where x_A , y_A , and z_A denote the principal axes of the water moiety (see below). Finally, for the molecular masses, we use 18.0103 and 20.0228 amu for H₂O and D₂O, respectively.

For both methods, we also require a PES describing the water–CO₂ interaction. We use the one obtained from the 12D water–CO₂ two-body PES computed by Wang and Bowman¹¹ for the fixed monomer geometries detailed above. We did make one minor modification to the resulting 5D PES. Since we found evidence for an unphysical “hole” (e.g., see Ref. 25) in the Wang and Bowman function in regions corresponding to small inter-monomer distances, we set the interaction PES to a large value (0.01 hartree) for all dimer geometries at which one (or both) of the O(water)-to-O(CO₂) internuclear distances is smaller than 1.5 Å. This fix is justifiable physically and eliminates any influence of the “hole” on our results.

B. The Brocks-type approach

1. Coordinates and Hamiltonian

We define, as specified by Brocks *et al.*,²⁰ a DF frame whose origin is at the c.m. of the dimer, whose axes are rotated by the Euler angles $\Omega \equiv (\alpha, \beta)$ from a space-fixed (SF) frame and whose z axis (i.e., $\hat{z}_D$) is parallel to the vector $\mathbf{r}_0$ pointing from the c.m. of monomer A (the water moiety) to that of monomer B (the CO₂) moiety. Second, we define the monomer-fixed frame for the water (MF_A) as the principal axis frame of that moiety, with $\hat{z}_A$ along the C₂ symmetry axis pointing toward the O nucleus, $\hat{y}_A$ as the out-of-plane principal axis parallel to the vector defined by the cross product of the O-to-H1 bond vector and the O-to-H2 bond vector, and $\hat{x}_A = \hat{y}_A \times \hat{z}_A$. Third, we define the z axis fixed to the CO₂ (i.e., $\hat{z}_B$) as the unit vector pointing from the C nucleus of that moiety to the O1 nucleus. Finally, we define the coordinates (in addition to α and β defined above): (i) $r_0 \equiv |\mathbf{r}_0|$, (ii) $\omega_A \equiv (\alpha_A, \beta_A, \gamma_A)$, the Euler angles that define the orientation of MF_A with respect to DF, and (iii) $\omega_B \equiv (\alpha_B, \beta_B)$, the Euler angles that define the orientation of $\hat{z}_B$ with respect to DF.

The intermolecular rovibrational Hamiltonian can now be written as

$$\hat{H}_{\text{inter}}(\mathbf{Q}, \Omega) = \hat{T}_{\text{inter}}(\mathbf{Q}, \Omega) + \hat{T}_{\text{rot}}^A(\omega_A) + \hat{T}_{\text{rot}}^B(\omega_B) + V_{\text{inter}}(\tilde{\mathbf{Q}}), \quad (1)$$

where $\mathbf{Q} \equiv (r_0, \omega_A, \omega_B)$ and $\tilde{\mathbf{Q}} \equiv (r_0, \tilde{\alpha}, \beta_A, \gamma_A, \beta_B)$ with $\tilde{\alpha} \equiv \alpha_A - \alpha_B$. The intermolecular kinetic-energy operator, $\hat{T}_{\text{inter}}$, is that given by Brocks *et al.*,²⁰

$$\hat{T}_{\text{inter}} \equiv -\frac{1}{2\mu_D} \frac{\partial^2}{\partial r_0^2} + \frac{1}{2\mu_D r_0^2} \left[\mathbf{J}^2 - \cot \beta \frac{\partial}{\partial \beta} + (\mathbf{j}^A + \mathbf{j}^B)^2 - 2\mathbf{J} \cdot (\mathbf{j}^A + \mathbf{j}^B) \right], \quad (2)$$

where μ_D is the reduced mass of the dimer, $\mathbf{J}$ is the vector operator associated with the rotational angular momentum of the dimer measured along the DF axes, and $\mathbf{j}^A$ and $\mathbf{j}^B$ are the vector operators

associated with the rotational angular momenta of the water and CO₂ moieties, respectively, as measured along the DF axes,

$$\hat{T}_{\text{rot}}^A \equiv \sum_{i_A} B_{i_A} (\hat{j}_{i_A}^A)^2, \quad (3)$$

where i_A is an index that runs over the principal axes of the water and $\hat{j}_{i_A}^A$ is the operator associated with the rotational angular momentum of the water moiety along axis i_A , is the rigid-rotor rotational kinetic-energy operator of the water monomer. Similarly,

$$\hat{T}_{\text{rot}}^B \equiv B (\hat{j}^B)^2 \quad (4)$$

is the rigid-rotor rotational kinetic-energy operator of the CO₂ monomer. Finally, $V_{\text{inter}}(\mathbf{Q})$ is the 5D intermolecular potential function from Wang and Bowman¹¹ referred above.

We note that the integration volume element associated with this $\hat{H}_{\text{inter}}$ is of the Wilson type:²⁶ $dr_0 d\omega_A d\omega_B d\Omega$. This choice facilitates the construction of the potential-optimized discrete variable representation (PODVR) covering the r_0 coordinate (see below).

2. Basis states

To diagonalize $\hat{H}_{\text{inter}}$, we use “uncoupled” basis functions of the Brock type analogous to those employed by Carrington *et al.* in diagonalizing Brock-type Hamiltonians in other dimer systems (e.g., see Ref. 27). These are of the form

$$|s, j_A, k_A, m, j_B; J, K\rangle \equiv |r_{0,s}\rangle |j_A, k_A, m\rangle |j_B, K - m\rangle |J, K\rangle, \quad (5)$$

where the $|r_{0,s}\rangle$ ($s = 1, \dots, N_{r_0}$) are potential-optimized DVR (PODVR)^{28,29} functions covering the r_0 coordinate (constructed by a procedure analogous to that described in Sec. 3.1.2 of Ref. 14), the $|j_A, k_A, m\rangle$ ($j_A = 0, 1, \dots, j_A^{\text{max}}$) are symmetric-top rotational eigenfunctions dependent on the ω_A coordinates, the $|j_B, -m\rangle$ ($j_B = 0, 1, \dots, j_B^{\text{max}}$) are spherical harmonics dependent on the ω_B coordinates, and the $|J, K\rangle$ ($K = -J, -J + 1, \dots, J$) are normalized “little- d ” Wigner matrix elements of the form

$$|J, K\rangle \equiv \sqrt{\frac{2J+1}{2}} d_{0,K}^J(\beta). \quad (6)$$

The full basis for a given value of J (J is a good quantum number, so the matrix of $\hat{H}_{\text{inter}}$ is block-diagonal in J) consists of all those functions of the type in Eq. (5) whose k_A and m values are allowed, given the values of j_A and j_B . Such a basis is also readily sorted into states that are symmetric or antisymmetric with respect to inversion, E^* , and H-nuclei-interchange, (H1, H2), symmetry operations that leave $\hat{H}_{\text{inter}}$ invariant. We make use of such sorting to block diagonalize the $\hat{H}_{\text{inter}}$ matrix with respect to E^* and (H1, H2) eigenvalues, as well. Note that a third symmetry operation—the interchange of the O nuclei of the CO₂, (O1, O2)—also leaves $\hat{H}_{\text{inter}}$ unchanged. We did not make use of this symmetry in factoring the $\hat{H}_{\text{inter}}$ matrix. However, we did analyze the computed eigenvectors as to their transformation properties with respect to this operation. In order to do this, we made use of the fact that the basis functions in Eq. (5) transform as

$$(O1, O2)|s, j_A, k_A, m, j_B; J, K\rangle = (-1)^{j_B} |s, j_A, k_A, m, j_B; J, K\rangle \quad (7)$$

so that any given eigenfunction, $|\psi\rangle$, of $\hat{H}_{\text{inter}}$ transforms as

$$(O1, O2)|\psi\rangle = \sum_{\Gamma} (-1)^{j_B} |s, j_A, k_A, m, j_B; J, K\rangle \times \langle s, j_A, k_A, m, j_B; JK|\psi\rangle, \quad (8)$$

where Γ represents the set of indices (s, j_A, k_A, m, j_B, K) .

All the bases that we used had $N_{r_0} = 20$ and $j_B^{\text{max}} = 32$. For the H₂O isotopologue, we used $j_A^{\text{max}} = 9$, while for the D₂O species, we used $j_A^{\text{max}} = 11$. We settled on these basis-set parameters after varying each of them over a range of values and performing convergence tests. For these parameters and $J = 0$, the numbers of basis states, absent any symmetry sorting, are 778 800 and 1 312 080 for H₂O–CO₂ and D₂O–CO₂, respectively. The corresponding numbers for $J = 1$ are 2 332 400 and 3 930 480, and for $J = 2$, they are 3 874 080 and 6 531 680. The number of independent basis states corresponding to the symmetry-sorted blocks is about a factor of four smaller than those listed above.

3. Diagonalization of $\hat{H}_{\text{inter}}$

We solve for the eigenvalues and eigenvectors of $\hat{H}_{\text{inter}}$ for a given value of J and for specific eigenvalues of the E^* and (H1, H2) operators by using the Chebyshev version³⁰ of filter-diagonalization³¹ (CFD). To compute the repeated effect of $\hat{H}_{\text{inter}}$ on a random initial state function, as required by this method, we divide such operation into two steps. In the first, we operate with the kinetic-energy portion of $\hat{H}_{\text{inter}}$ by using direct matrix-on-vector multiplication. The kinetic-energy matrix elements in the basis in Eq. (5) are readily evaluated analytically or, in the case of r_0 -dependent terms, by numerical transformation from analytically determined matrix elements in an r_0 -dependent, particle-in-a-box-eigenfunction basis. In the second step, operation on the state function with the potential energy, we transform that the function from the basis representation to a 5D quadrature-grid representation (over the $\mathbf{Q}$ coordinates) by well-known operations (e.g., see Ref. 12), then multiply the state function at each grid point by the value of V at that point, and, finally, transform the result back to the basis representation to complete the operation. The results of the two steps are added to yield the desired result. During the course of repeated $\hat{H}_{\text{inter}}$ operations, energy-filtered basis functions are built up in the CFD algorithm. At the end of the Chebyshev propagation, the matrix of $\hat{H}_{\text{inter}}$ in this energy-filtered basis is computed and then diagonalized to yield eigenvectors and eigenvalues in the desired energy window of the spectrum of the Hamiltonian. In this paper, the relevant windows invariably correspond to the lowest-energy part of each spectrum.

C. A dimer-fixed frame embedded in the CO₂ moiety

1. Coordinates and Hamiltonian

The Brocks-type approach has some drawbacks in calculations of the intermolecular rovibrational states of a heavy-light weakly bound dimer like CO₂–water. First, the angular basis for the heavy species must include a relatively large number of functions in order to achieve convergence (e.g., in the above $j_B^{\text{max}} = 32$). Second, the Brocks coordinate system does not lend itself to ready physical interpretation/visualization of many of the computed intermolecular eigenfunctions. As such, we have also computed here the intermolecular rovibrational states of the two water–CO₂ isotopologues by using a Hamiltonian (derived here) expressed in terms

of coordinates referred to a DF frame attached to the CO₂ (heavy) moiety.

We define this DF frame (which we denote DF') as that which is obtained by rotating a space-fixed frame through the two Euler angles (α' , β') to yield a new frame whose $\hat{z}$ axis (i.e., $\hat{z}'_D$) is parallel to the vector from the C nucleus to the O1 nucleus of CO₂. The DF' origin is at the c.m. of the dimer. In this frame, the intermolecular rovibrational kinetic-energy operator can be obtained by starting from Eq. (3.11) of Ref. 32, a general expression for the kinetic-energy operator of a dimer composed of two rigid monomers with the dimer frame fixed to one of the monomers. We write that equation in the form

$$\hat{T} = -\frac{\nabla_d^2}{2\mu_D} + \sum_{i'_D} \frac{(\hat{J}_{i'_D} - \hat{L}_{i'_D} - \hat{j}_{i'_D})^2}{2I_{i'_D}} + \sum_{i_A} \frac{\hat{J}_{i_A}^2}{2I_{i_A}}. \quad (9)$$

In this expression, μ_D is the dimer reduced mass, ∇_d^2 is the Laplacian associated with $\mathbf{d}$, the vector from the CO₂ c.m. to that of the water moiety (i.e., $\mathbf{d} \equiv -\mathbf{r}_0$), i'_D is an index that runs over the three axes of the DF' frame, $\hat{J}_{i'_D}$ is the operator associated with the component of the total angular momentum of the dimer (apart from spin) along the i'_D axis, $\hat{L}_{i'_D}$ is the operator associated with the component along the i'_D axis of the orbital angular momentum due to the rotation of the monomer c.m.'s about the dimer c.m., $\hat{j}_{i'_D}$ is the operator associated with the rotational angular momentum of the water moiety as measured along the i'_D direction, $I_{i'_D}$ is the principal moment of inertia of the CO₂ moiety about the i'_D axis, i_A is an index that runs over the three principal axes of the water moiety, $\hat{J}_{i_A}$ is the operator associated with the rotational angular momentum of the water moiety measured along the i_A axis, and I_{i_A} is the principal moment of inertia of the water moiety about the i_A axis.

Now, $I_{z'_D} = 0$ for linear CO₂. Hence, for Eq. (9) to be defined, it must be that the z'_D component of $\mathbf{J}$ equals the sum of the z'_D components of $\mathbf{L}$ and $\mathbf{j}$. (This condition can be guaranteed by proper choice of basis—see below.) In that case,

$$\hat{T} = -\frac{\nabla_d^2}{2\mu_D} + \frac{1}{2I} [(\hat{J}_{x'_D} - \hat{L}_{x'_D} - \hat{j}_{x'_D})^2 + (\hat{J}_{y'_D} - \hat{L}_{y'_D} - \hat{j}_{y'_D})^2] + \sum_{i_A} \frac{\hat{J}_{i_A}^2}{2I_{i_A}}, \quad (10)$$

where I is the moment of inertia of linear CO₂. We rewrite this equation in terms of the raising and lowering operators as

$$\hat{J}_{\pm} \equiv \hat{J}_{x'_D} \mp i\hat{J}_{y'_D}, \quad \hat{L}_{\pm} \equiv \hat{L}_{x'_D} \pm i\hat{L}_{y'_D}, \quad \hat{j}_{\pm} \equiv \hat{j}_{x'_D} \pm i\hat{j}_{y'_D} \quad (11)$$

to obtain

$$\hat{T} = -\frac{\nabla_d^2}{2\mu_D} + B[(\hat{j}^2 - \hat{j}_{z'_D}^2) + (\hat{L}^2 - \hat{L}_{z'_D}^2) + (\hat{J}^2 - \hat{J}_{z'_D}^2) - \hat{J}_+(\hat{L}_+ + \hat{j}_+) - \hat{J}_-(\hat{L}_- + \hat{j}_-) + \hat{L}_+\hat{j}_- + \hat{L}_-\hat{j}_+] + \sum_{i_A} B_{i_A} \hat{J}_{i_A}^2, \quad (12)$$

where $B \equiv 1/(2I)$ is the rotational constant of the rigid CO₂ and B_{i_A} are the rotational constants of the rigid water and where we have used the facts that (a) all the components of $\hat{\mathbf{J}}$ commute with those of $\hat{\mathbf{L}}$ and $\hat{\mathbf{j}}$ and (b) all the components of $\hat{\mathbf{L}}$ commute with those of $\hat{\mathbf{j}}$. It

is convenient to re-express this operator as the sum of a vibrational operator and a rovibrational one,

$$\hat{T} = \hat{T}_{vib} + \hat{T}_{rv}, \quad (13)$$

where

$$\hat{T}_{vib} \equiv -\frac{\nabla_d^2}{2\mu_D} + B[(\hat{L}^2 - \hat{L}_{z'_D}^2) + (\hat{j}^2 - \hat{j}_{z'_D}^2) + \hat{L}_+\hat{j}_- + \hat{L}_-\hat{j}_+] + \sum_{i_A} B_{i_A} \hat{J}_{i_A}^2 \quad (14)$$

and

$$\hat{T}_{rv} \equiv B[(\hat{j}^2 - \hat{j}_{z'_D}^2) - \hat{J}_+(\hat{L}_+ + \hat{j}_+) - \hat{J}_-(\hat{L}_- + \hat{j}_-)]. \quad (15)$$

It remains now to choose specific coordinates. We already have the Euler angles (α' , β') fixing the DF' frame relative to a space-fixed frame. To fix the orientation of the water moiety's principal axis system relative to DF', we use the Euler angles (α'_A , β'_A , γ'_A). Finally, for the components of $\mathbf{d}$, we use the cylindrical coordinates (ρ , ϕ , z) defined such that

$$d_{x'_D} = \rho \cos \phi, \quad d_{y'_D} = \rho \sin \phi, \quad d_{z'_D} = z. \quad (16)$$

Note that in this set of coordinates, the intermolecular PES does not depend on α'_A and ϕ individually but on their difference. Since the PES is also independent of α' and β' , the number of intermolecular vibrational coordinates is five, as must be the case for a dimer composed of a rigid linear species and a rigid nonlinear one.

With Eq. (16), the chain rule, and the definitions of the $\hat{L}_{i'_D}$ in terms of $d_{x'_D}$, $d_{y'_D}$, and $d_{z'_D}$, one can determine the explicit dependences of $\hat{T}_{vib}$ and $\hat{T}_{rv}$ on the cylindrical coordinates. For $\hat{T}_{vib}$, one has

$$\begin{aligned} \hat{T}_{vib} = & -\left(\frac{1}{2\mu_D} + Bz^2\right) \left(\frac{\partial^2}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial}{\partial \rho} - \frac{\hat{L}_{z'_D}^2}{\rho^2}\right) \\ & - \left(\frac{1}{2\mu_D} + B\rho^2\right) \frac{\partial^2}{\partial z^2} + 2Bz \frac{\partial}{\partial z} + B\left(\rho \frac{\partial}{\partial \rho}\right) \left(1 + 2z \frac{\partial}{\partial z}\right) \\ & + B(\hat{j}_+ e^{-i\phi} - \hat{j}_- e^{+i\phi}) \left(\rho \frac{\partial}{\partial z} - z \frac{\partial}{\partial \rho}\right) \\ & - B(\hat{j}_+ e^{-i\phi} + \hat{j}_- e^{+i\phi}) \frac{z \hat{L}_{z'_D}}{\rho} + B(\hat{j}^2 - \hat{j}_{z'_D}^2) + \sum_{i_A} B_{i_A} \hat{J}_{i_A}^2, \end{aligned} \quad (17)$$

where $\hat{L}_{z'_D} = -i(\partial/\partial\phi)$. For $\hat{T}_{rv}$, one substitutes for $\hat{L}_{\pm}$ in Eq. (15) the following equation:

$$\hat{L}_{\pm} = e^{\pm i\phi} \left[\mp \rho \frac{\partial}{\partial z} \pm z \frac{\partial}{\partial \rho} - \frac{z}{\rho} \hat{L}_{z'_D} \right]. \quad (18)$$

Finally, note that the integration volume pertaining to the $\hat{T}_{vib}$ and $\hat{T}_{rv}$ operators presented immediately above goes as $\rho d\rho d\phi dz$. It is desirable to transform these operators so that a Wilson-type integration volume,²⁶ proportional to $d\rho d\phi dz$, is obtained. For operator $\hat{O}$, this is accomplished by the transformation $\hat{O}^{(W)} = \rho^{1/2} \hat{O} \rho^{-1/2}$. Such a transformation is readily affected by making the following substitutions into the expressions for $\hat{T}_{vib}$ and $\hat{T}_{rv}$:

$$\frac{\partial}{\partial \rho} \rightarrow \left[\frac{\partial}{\partial \rho} - \frac{1}{2\rho} \right] \quad \text{and} \quad \left[\frac{\partial^2}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial}{\partial \rho} \right] \rightarrow \left[\frac{\partial^2}{\partial \rho^2} + \frac{1}{4\rho^2} \right]. \quad (19)$$

This yields

$$\begin{aligned}\hat{T}_{vib}^{(W)} = & -\left(\frac{1}{2\mu_D} + Bz^2\right)\left(\frac{\partial^2}{\partial\rho^2} + \frac{1}{4\rho^2} - \frac{\hat{L}_{z'_D}^2}{\rho^2}\right) - \left(\frac{1}{2\mu_D} + B\rho^2\right) \\ & \times \frac{\partial^2}{\partial z^2} + 2Bz\frac{\partial}{\partial z} + B\left(\rho\frac{\partial}{\partial\rho} - \frac{1}{2}\right)\left(1 + 2z\frac{\partial}{\partial z}\right) \\ & + B\left(\hat{j}_+e^{-i\phi} - \hat{j}_-e^{+i\phi}\right)\left(\rho\frac{\partial}{\partial z} - z\frac{\partial}{\partial\rho} + \frac{z}{2\rho}\right) \\ & - B\left(\hat{j}_+e^{-i\phi} + \hat{j}_-e^{+i\phi}\right)\frac{z\hat{L}_{z'_D}}{\rho} + B(\hat{j}^2 - \hat{j}_{z'_D}^2) \\ & + \sum_{i_A} B_{i_A} \hat{j}_{i_A}^2.\end{aligned}\quad (20)$$

$\hat{T}_{rv}^{(W)}$ is obtained by substituting

$$\hat{L}_{\pm}^{(W)} = e^{\pm i\phi} \left[\mp \rho \frac{\partial}{\partial z} \pm z \frac{\partial}{\partial \rho} - \frac{z}{\rho} \left(\hat{L}_{z'_D} \pm \frac{1}{2} \right) \right] \quad (21)$$

for $\hat{L}_{\pm}$ in Eq. (15).

The full intermolecular rovibrational Hamiltonian is finally given by

$$\hat{H}_{\text{inter}}' = \hat{T}_{vib}^{(W)} + \hat{T}_{rv}^{(W)} + V_{\text{inter}}(\tilde{\mathbf{Q}}'), \quad (22)$$

where V_{inter} is the Wang and Bowman PES function described in Sec. II A and $\tilde{\mathbf{Q}}' \equiv (\rho, \tilde{\phi}, z, \beta'_A, \gamma'_A)$ with $\tilde{\phi} \equiv \alpha'_A - \phi$.

2. Basis states

To diagonalize $\hat{H}_{\text{inter}}'$, we use a primitive basis composed of functions of the form

$$|t, u, j_A, k_A, m; J, K\rangle \equiv |\rho_t\rangle |z_u\rangle \frac{e^{i(K-m)\phi}}{\sqrt{2\pi}} |j_A, k_A, m\rangle |J, K\rangle, \quad (23)$$

where the $|\rho_t\rangle$ ($t = 1, \dots, N_\rho$) constitute a PODVR covering the ρ coordinate, the $|z_u\rangle$ ($u = 1, \dots, N_z$) constitute a PODVR covering the z coordinate, the $|j_A, k_A, m\rangle$ ($j_A = 0, \dots, j_A^{\text{max}}$) here are symmetric-top eigenfunctions covering the $(\alpha'_A, \beta'_A, \gamma'_A)$ coordinates, and the $|J, K\rangle$ here are little- d Wigner matrix elements of the form $|J, K\rangle \equiv [(2J+1)/2]^{1/2} d_{0,K}^J(\beta')$. [Note that these basis functions are all eigenfunctions of $\hat{J}_{z'_D}$, $\hat{L}_{z'_D}$, and $\hat{j}_{z'_D}$ with eigenvalues equal to K , $K-m$, and m , respectively, so that matrix elements of $(\hat{J}_{z'_D} - \hat{L}_{z'_D} - \hat{j}_{z'_D})$ in the basis in Eq. (23) always vanish and the matrix elements of the operator in Eq. (9) are always defined.]

The PODVRs covering the ρ and z coordinates were constructed as follows: To obtain the $|\rho_t\rangle$ for a given isotopologue, the eigenvalue equation

$$\left[-\left(\frac{1}{2\mu_D} + Bz_{eq}^2\right) \frac{\partial^2}{\partial\rho^2} + V_{ad}^\rho(\rho) \right] \psi(\rho) = \epsilon \psi(\rho), \quad (24)$$

where $z_{eq} = 0$ and $V_{ad}^\rho(\rho)$ is the 1D potential obtained by minimizing V_{inter} with respect to $(z, \tilde{\phi}, \beta'_A, \gamma'_A)$ as a function of ρ , was solved in a basis of 100 sinc-DVR functions obtained from particle-in-a-box eigenfunctions corresponding to a box extending from $\rho = 2.275$ to 8.626 Å. The matrix of ρ in a basis of the resulting N_ρ lowest-energy

eigenstates was then constructed and diagonalized, yielding as eigenvalues the ρ_t quadrature points and, ultimately, the transformation matrix between the $|\rho_t\rangle$ functions and the original particle-in-a-box functions. Similarly, to obtain the $|z_u\rangle$ for an isotopologue, the eigenvalue equation

$$\left[-\left(\frac{1}{2\mu_D} + B\rho_{eq}^2\right) \frac{\partial^2}{\partial z^2} + V_{ad}^z(z) \right] \psi(z) = \epsilon \psi(z), \quad (25)$$

where $\rho_{eq} = 2.856$ Å and $V_{ad}^z(z)$ is the 1D potential obtained by minimizing V_{inter} with respect to $(\rho, \tilde{\phi}, \beta'_A, \gamma'_A)$ as a function of z , was solved in a basis of 100 sinc-DVR functions obtained from particle-in-a-box eigenfunctions corresponding to a box extending from $z = -1.852$ to $+1.852$ Å. The matrix of z in a basis of the resulting N_z lowest-energy eigenstates was then constructed and diagonalized to yield the z_u quadrature points and, ultimately, the transformation matrix between the $|z_u\rangle$ functions and the original particle-in-a-box functions.

The basis sets of the type in Eq. (23) that we employed in diagonalizing $\hat{H}_{\text{inter}}'$ had $j_A^{\text{max}} = 9$ and 11 for the $\text{H}_2\text{O}-\text{CO}_2$ and $\text{D}_2\text{O}-\text{CO}_2$ dimers, respectively. States with all allowed values of k_A and m for a given j_A were included in each basis. For both isotopologues, $N_\rho = 16$ and $N_z = 16$. With these parameters, the total number of $J = 0$ basis states equaled 340 480 and 588 800 for the H_2O and D_2O dimers, respectively. The corresponding numbers for $J = 1$ and $J = 2$ are, respectively, about a factor of 3 and 5 larger.

It should be emphasized that these basis sets yield the same degree of convergence (to within a few thousandths of cm^{-1}) as that characterizing the results from the Brocks-type approach described in Sec. II B, although their sizes are less than a half of the corresponding bases sizes employed in the Brocks-type approach. This very significant reduction in the basis-set size afforded by the new approach would be of particularly great importance in the case that the calculations are extended to flexible monomers in 12D, given how expensive the evaluations of the PES are.

Note that we did not attempt to symmetry-sort these bases, as we did in the Brocks-type calculations—we were primarily interested in checking the results of the new method against those obtained by the Brocks approach. Nevertheless, such symmetry sorting and the block diagonalization of the $\hat{H}_{\text{inter}}'$ matrix in the basis in Eq. (23) should be quite straightforward, given the transformation properties

$$E^* |t, u, j_A, k_A, m; J, K\rangle = (-1)^{J+K+m+k_A} |t, u, j_A, \bar{k}_A, \bar{m}; J, \bar{K}\rangle, \quad (26)$$

$$(H1, H2) |t, u, j_A, k_A, m; J, K\rangle = (-1)^{k_A} |t, u, j_A, k_A, m; J, K\rangle, \quad (27)$$

and

$$(O1, O2) |t, u, j_A, k_A, m; J, K\rangle = (-1)^{J+j_A} |t, \bar{u}, j_A, k_A, \bar{m}; J, \bar{K}\rangle, \quad (28)$$

where $\bar{K} \equiv -K$, $\bar{m} \equiv -m$, $\bar{k}_A = -k_A$, and $|z_u\rangle$ represents the z PODVR function centered at the quadrature point that is the negative of that corresponding to $|z_u\rangle$. In fact, we did make use of Eqs. (26)–(28) to analyze the symmetry properties of the eigenvectors computed for $\hat{H}_{\text{inter}}'$.

3. Diagonalization of $\hat{H}'_{\text{inter}}$

We diagonalized $\hat{H}'_{\text{inter}}$ by a procedure very similar to that described in Sec. II B 3. Again, we used the Chebyshev version of filter diagonalization. We operated on state functions expressed in the basis in Eq. (23) with the kinetic-energy portion of $\hat{H}'_{\text{inter}}$ by using direct matrix-vector operation. The portions of the matrix elements involving the angular coordinates were evaluated analytically. Those involving ρ and z were obtained by numerically transforming to the PODVR representations matrix elements evaluated analytically in the particle-in-a-box representations from which the PODVRs were computed. Operation with the potential energy was affected by transforming the state function of interest to a 5D quadrature grid covering the $\hat{Q}'$ coordinates by standard methods. The state function in the grid representation was multiplied by the value of V_{inter} at each point on the grid, and the result was transformed back to the basis representation. The remainder of the filter diagonalization procedure proceeded as described in Sec. II B 3.

III. RESULTS AND DISCUSSION

A. Rotation-tunneling states and rotational constants in the vibrational ground states of the complexes

The low-energy rotational states of the $\text{H}_2\text{O}-\text{CO}_2$ and $\text{D}_2\text{O}-\text{CO}_2$ complexes in their ground intermolecular vibrational states from the quantum 5D calculations are given in Tables I and II, respectively. They are referred to as the rotation-tunneling states since each rovibrational level of the complex is split into a doublet due to the internal rotation of the two moieties around the

TABLE I. Rotation-tunneling states in the vibrational ground state of $\text{H}_2\text{O}-\text{CO}_2$ from quantum 5D rigid-monomer calculations.

Assignment ^a	ΔE (cm ⁻¹)	Irrep ^b	Tunneling splitting (cm ⁻¹) (MHz)
0 ₀₀ (a)	0.0000	A' ₁	
0 ₀₀ (b)*	0.1447	A'' ₁	0.1447 (4338.6)
1 ₀₁ (a)	0.2661	A' ₂	
1 ₀₁ (b)*	0.4109	A'' ₂	0.1448 (4339.8)
1 ₁₁ (a)*	0.4946	B' ₁	
1 ₁₁ (b)	0.6381	B'' ₁	0.1435 (4302.4)
1 ₁₀ (a)*	0.5402	B' ₂	
1 ₁₀ (b)	0.6838	B'' ₂	0.1435 (4303.0)
2 ₀₂ (a)	0.7922	A' ₁	
2 ₀₂ (b)*	0.9370	A'' ₁	0.1448 (4341.7)
2 ₁₂ (a)*	0.9812	B' ₂	
2 ₁₂ (b)	1.1247	B'' ₂	0.1435 (4303.2)
2 ₁₁ (a)*	1.1181	B' ₁	
2 ₁₁ (b)	1.2617	B'' ₁	0.1436 (4305.1)
2 ₂₁ (a)	1.8021	A' ₂	
2 ₂₁ (b)*	1.9450	A'' ₂	0.1429 (4282.9)
2 ₂₀ (a)	1.8083	A' ₁	
2 ₂₀ (b)*	1.9512	A'' ₁	0.1429 (4283.3)

^aThe states whose assignments are marked with an asterisk do not exist in nature as they are antisymmetric with respect to interchange of the ¹⁶O nuclei.

^bIrreducible representation of the group G_8 . See, for example, Ref. 3.

TABLE II. Rotation-tunneling states in the vibrational ground state of $\text{D}_2\text{O}-\text{CO}_2$ from quantum 5D rigid-monomer calculations.

Assignment ^a	ΔE (cm ⁻¹)	Irrep ^b	Tunneling splitting (cm ⁻¹) (MHz)
0 ₀₀ (a)	0.0000	A' ₁	
0 ₀₀ (b)*	0.0039	A'' ₁	0.0039 (117.0)
1 ₀₁ (a)	0.2425	A' ₂	
1 ₀₁ (b)*	0.2464	A'' ₂	0.0039 (117.2)
1 ₁₁ (a)*	0.4766	B' ₁	
1 ₁₁ (b)	0.4793	B'' ₁	0.0028 (83.4)
1 ₁₀ (a)*	0.5155	B' ₂	
1 ₁₀ (b)	0.5183	B'' ₂	0.0028 (83.6)
2 ₀₂ (a)	0.7230	A' ₁	
2 ₀₂ (b)*	0.7269	A'' ₁	0.0039 (117.9)
2 ₁₂ (a)*	0.9226	B' ₂	
2 ₁₂ (b)	0.9254	B'' ₂	0.0028 (82.6)
2 ₁₁ (a)*	1.0394	B' ₁	
2 ₁₁ (b)	1.0422	B'' ₁	0.0028 (83.5)
2 ₂₁ (a)	1.7395	A' ₂	
2 ₂₁ (b)*	1.7432	A'' ₂	0.0038 (112.8)
2 ₂₀ (a)	1.7439	A' ₁	
2 ₂₀ (b)*	1.7477	A'' ₁	0.0038 (112.8)

^aThe states whose assignments are marked with an asterisk do not exist in nature as they are antisymmetric with respect to interchange of the ¹⁶O nuclei.

^bIrreducible representation of the group G_8 . See, for example, Ref. 3.

C··O vdW bond. However, the two equivalent ¹⁶O nuclei of CO_2 have zero nuclear spin and are therefore bosons. Consequently, the total wavefunction of the complex must be symmetric with respect to their interchange. Detailed considerations of the wavefunction symmetry and nuclear-spin statistics^{1,2} lead to the conclusion that for each level, only one component of the tunneling doublet meets this requirement and is allowed, while the other component does not exist in nature. $\text{H}_2\text{O}/\text{D}_2\text{O}-\text{CO}_2$ is an asymmetric top and its rotational levels are labeled as $J_{K_a K_c}$, where J is the total angular momentum quantum number and K_a and K_c indicate the K values (which are less than or equal to J) of the prolate and oblate levels to which the level correlates.³³

The level energies of $\text{H}_2\text{O}-\text{CO}_2$ in Table I and $\text{D}_2\text{O}-\text{CO}_2$ in Table II are measured from their quantum 5D ground-state energies at -768.64 and -814.20 cm⁻¹, respectively, relative to the separated rigid monomers. For comparison, full-dimensional diffusion Monte Carlo calculations of the $\text{H}_2\text{O}-\text{CO}_2$ complex on the same (12D) PES gave for the ground-state energy the value of -787 cm⁻¹ relative to the ground state energies of the separated flexible monomers.¹¹ This is ~ 18 cm⁻¹, or 2.3%, lower than the quantum 5D result. The difference is mostly likely caused by the assumption of rigid monomers in the quantum 5D treatment.

The calculated internal rotation tunneling splitting of 0.1447 cm⁻¹ for the lowest rotational state 0₀₀ of $\text{H}_2\text{O}-\text{CO}_2$ in Table I is in very good agreement, to within 6%, with the corresponding experimentally determined tunneling splitting of $0.1356(13)$ cm⁻¹.² The remaining small difference between the calculated and experimental splitting can have several causes, including slight inaccuracy of the PES employed and the rigid-monomer

approximation. Another possible contribution to this minor discrepancy can come from the experiment. The experimental value of the tunneling splitting was derived from the near-IR spectrum of the $\text{H}_2\text{O}-\text{CO}_2$ complex in the region of the asymmetric O–H stretch band of the water monomer under the assumption that the tunneling splitting is independent of the intramolecular vibrational excitation.² A more recent spectroscopic study of the same complex in the ν_2 bend region of the H_2O monomer⁶ yielded the tunneling splitting about 5% smaller than that obtained previously for the O–H stretch excited $\text{H}_2\text{O}-\text{CO}_2$.² This indicates a weak but measurable dependence of the splitting on the intramolecular vibrational excitation of the water moiety.

In the case of $\text{D}_2\text{O}-\text{CO}_2$, Table II shows for the lowest rotational state 0_{00} the calculated tunneling splitting owing to internal rotation of 0.0039 cm^{-1} , which is, as expected, much smaller than the splitting for $\text{H}_2\text{O}-\text{CO}_2$.

A closer look at Tables I and II leads to an interesting observation that states with the same K_a value have the same splitting, independent of K_c , whereas the splittings are slightly different for

the states with different K_a 's. The variation is somewhat more pronounced for $\text{D}_2\text{O}-\text{CO}_2$, for which the splittings in Table II are 0.0039 cm^{-1} for the states with $K_a = 0$ and 0.0028 cm^{-1} for the $K_a = 1$ states.

Table III lists the rotational transition frequencies calculated for the $\text{H}_2\text{O}-\text{CO}_2$ and $\text{D}_2\text{O}-\text{CO}_2$ complexes using the energies of the states in Tables I and II, respectively. Also given are the corresponding experimental values by Peterson and Klemperer.¹ The observed rotational transitions reported in Ref. 3 are nearly identical and could have been used instead for this purpose. A comparison of the computed and measured rotational transition frequencies reveals their outstanding agreement. The theory and experiment differ by at most 1 MHz for $\text{H}_2\text{O}-\text{CO}_2$ and no more than 10 MHz for $\text{D}_2\text{O}-\text{CO}_2$.

The calculated and experimentally derived rotational constants A , B , and C of $\text{H}_2\text{O}-\text{CO}_2$ and $\text{D}_2\text{O}-\text{CO}_2$ complexes in their ground vibrational states are presented in Table IV. It is easy to see that the agreement between the two sets of values is extraordinary. The differences do not exceed 0.2% for $\text{H}_2\text{O}-\text{CO}_2$ and 0.06% for $\text{D}_2\text{O}-\text{CO}_2$. These results, together with those for the rotational transition frequencies in Table III, show that the PES employed, by Wang and Bowman,¹¹ describes the rotational properties of both vdW complexes with high accuracy. Since the rotational constants B and C have similar values, the two complexes can be viewed as near-prolate symmetric tops.

TABLE III. Calculated and observed rotational transition frequencies (in MHz) in the vibrational ground states of $\text{H}_2\text{O}-\text{CO}_2$ and $\text{D}_2\text{O}-\text{CO}_2$.

	Assignment	Calculated	Observed ^a
$\text{H}_2\text{O}-\text{CO}_2$	$1_{01}-0_{00}$	7 977.41	7 978.562 (10)
	$2_{02}-1_{01}$	15 771.19	15 771.229 (30)
	$2_{11}-1_{10}$	17 326.02	17 326.398 (15)
	$1_{10}-1_{11}$	1 369.14	1 369.455 (10)
	$2_{20}-2_{21}$	185.35	185.388 (2)
$\text{D}_2\text{O}-\text{CO}_2$	$1_{01}-0_{00}$	7 268.92	7 265.550 (50)
	$2_{02}-1_{01}$	14 405.88	14 396.900 (40)
	$2_{11}-1_{10}$	15 705.60	15 695.560 (100)
	$1_{10}-1_{11}$	1 167.18	1 166.360 (5)
	$2_{11}-2_{12}$	3 501.24	3 498.780 (20)
	$2_{20}-2_{21}$	133.90	133.662 (25)

^aFrom Ref. 1.

B. Intermolecular vibrational states and their tunneling shifts

Low-energy intermolecular vibrational states of $\text{D}_2\text{O}-\text{CO}_2$ and $\text{H}_2\text{O}-\text{CO}_2$ complexes up to about 200 cm^{-1} above the ground state, obtained from the quantum 5D rigid-monomer calculations, are presented in Tables V and VI, respectively. Their energies (ΔE) are relative to the quantum 5D ground-state energies, -814.20 cm^{-1} for $\text{D}_2\text{O}-\text{CO}_2$ and -768.64 cm^{-1} for $\text{H}_2\text{O}-\text{CO}_2$.

Also given next to the energy of each state in both tables is what we refer to as the internal-rotation tunneling shift (δ), defined as follows: Each rovibrational level of the complex is split by the internal rotation into a closely spaced pair of tunneling states. As outlined earlier, only one component of each tunneling doublet

TABLE IV. Calculated and experimentally derived rotational constants (in MHz) in the vibrational ground states of $\text{H}_2\text{O}-\text{CO}_2$ and $\text{D}_2\text{O}-\text{CO}_2$.

	$\text{H}_2\text{O}-\text{CO}_2$		$\text{D}_2\text{O}-\text{CO}_2$	
	Calculated ^a	Measured	Calculated ^a	Measured
$(B + C)\text{ (MHz)}^b$	7 978.0	7 979.0	7 269.06	7 265.624
$(B - C)\text{ (MHz)}^b$	1 353.8/1 368.7	1 370.3	1 167.04/1 166.93	1 167.290
$A - (B + C)/2\text{ (MHz)}^b$	7 523.2/7 515.8	7 517	7 584.46/7 584.49	7 589.371
$A\text{ (MHz)}^c$	11 512.3	11 501.23	11 219.0	11 220.12
$B\text{ (MHz)}^c$	4 665.9	4 675.078	4 218.0	4 216.473
$C\text{ (MHz)}^c$	3 312.1	3 304.617	3 051.0	3 049.182

^aWhere two calculated numbers are quoted, the first is derived from $J = 1$ levels and the second from $J = 2$ levels. Otherwise, the number is derived from $J = 1$ levels.

^bMeasured values from Ref. 1.

^cMeasured values from Ref. 3.

TABLE V. Low-energy intermolecular vibrational states of D₂O–CO₂ from quantum 5D rigid-monomer calculations. $\langle d \rangle$, Δd , and Δz (all in Å), as well as $\Delta \cos \beta'_A$ and $\Delta \sin \theta$, are defined in the text.

	ΔE^a (δ) ^b	$\langle d \rangle$ (Δd)	Δz	$\Delta \cos \beta'_A$	$\Delta \sin \theta$	Irrep ^c	Assignment
1	0.00 (−0.0019)	2.950 (0.111)	0.277	0.327	0.403	A_1'	Ground state
2	36.09 (−0.0003)	2.963 (0.113)	0.434	0.501	0.394	B_2''	ν_b
3	72.95 (−0.0025)	2.956 (0.113)	0.280	0.330	0.624	B_1''	ν_{inv}
4	75.81 (−0.0059)	2.985 (0.121)	0.553	0.581	0.389	A_1'	$2\nu_b$
5	86.80 (0.0633)	2.972 (0.114)	0.294	0.409	0.420	A_2'	ν_{ir}
6	97.47 (−0.0095)	3.010 (0.192)	0.293	0.340	0.405	A_1'	ν_s
7	112.99 (−0.0093)	3.015 (0.149)	0.649	0.624	0.389	B_2''	$3\nu_b$
8	116.85 (0.0150)	2.974 (0.115)	0.427	0.409	0.592	A_2'	$\nu_{\text{inv}} + \nu_b$
9	127.65 (−0.0459)	2.986 (0.117)	0.439	0.535	0.432	B_1''	$\nu_{\text{ir}} + \nu_b$
10	134.15 (−0.0006)	3.007 (0.152)	0.458	0.470	0.400	B_2''	$\nu_s + \nu_b$
11	140.99 (0.0281)	2.998 (0.145)	0.352	0.444	0.435	B_2''	ν_{rock}
12	147.95 (−0.0291)	3.044 (0.169)	0.743	0.664	0.388	A_1'	$4\nu_b$
13	156.54 (−0.0714)	2.969 (0.116)	0.305	0.394	0.665	A_1'	$2\nu_{\text{inv}}$
14	159.03 (−0.0140)	3.003 (0.140)	0.534	0.471	0.587	B_1''	$\nu_{\text{inv}} + 2\nu_b$
15	160.28 (0.0465)	2.981 (0.118)	0.306	0.418	0.554	B_2''	$\nu_{\text{inv}} + \nu_{\text{ir}}$
16	162.09 (−0.8538)	2.995 (0.119)	0.323	0.424	0.459	A_1'	$2\nu_{\text{ir}}$
17	168.32 (0.0900)	3.017 (0.146)	0.549	0.588	0.449	A_2'	$\nu_{\text{ir}} + 2\nu_b$
18	170.16 (−0.0229)	3.014 (0.181)	0.331	0.373	0.619	B_1''	$\nu_{\text{inv}} + \nu_s$
19	178.99 (−0.0904)	3.037 (0.164)	0.575	0.550	0.392	A_1'	$\nu_s + 2\nu_b$
20	181.09 (−0.0271)	3.081 (0.192)	0.850	0.704	0.389	B_2''	$5\nu_b$
21	182.00 (0.2591)	3.028 (0.181)	0.341	0.442	0.411	A_2'	$\nu_{\text{ir}} + \nu_s$
22	182.87 (−0.1465)	3.030 (0.181)	0.438	0.474	0.395	A_1'	$\nu_{\text{rock}} + \nu_b$
23	190.84 (−0.0743)	3.059 (0.222)	0.378	0.439	0.403	A_1'	$2\nu_s$
24	197.51 (0.0897)	3.031 (0.165)	0.612	0.498	0.570	A_2'	$\nu_{\text{inv}} + 3\nu_b$
25	201.55 (0.0034)	2.991 (0.118)	0.460	0.435	0.670	B_2''	$2\nu_{\text{inv}} + \nu_b$

^aEnergy in cm^{−1} relative to the ground-state energy of the complex.^bTunneling shift in cm^{−1}.^cIrreducible representation of the group G_8 . See, for example, Ref. 3.

satisfies the combined requirements imposed by the spatial wavefunction symmetry and nuclear-spin constraints and is the one listed in Table V for D₂O–CO₂ and Table VI for H₂O–CO₂. The other tunneling component is not physically realizable and, therefore, does not appear in these tables. The tunneling shifts in Tables V and VI are calculated relative to the average energy of each pair of tunneling states. If the state listed is the lower-energy component of the doublet, then its shift is equal to $-1/2$ of the tunneling splitting. If it is the higher-energy member of the pair, then its shift equals $+1/2$ of the tunneling splitting. Clearly, the tunneling splitting of a state is equal to $2|\delta|$.

Besides ΔE and δ , Tables V and VI display for each state $\langle d \rangle$ the expectation value of $d \equiv |\mathbf{d}|$, where $\mathbf{d}$ is the vector defined in Sec. II C, from the c.m. of CO₂ to that of the water moiety and the root-mean-square (rms) amplitude Δd . Also shown are Δz , $\Delta \cos \beta'_A$, and $\Delta \sin \theta$. The first two quantities are associated with the second approach described in Sec. II C: z is one of the cylindrical coordinates of $\mathbf{d}$ defined in Eq. (16) and is aligned with the internuclear axis of CO₂, while β'_A is the angle between the symmetry axis of the water moiety and the CO₂ internuclear axis (essentially the vdW bend angle). Finally, θ is the angle that the symmetry axis of water makes with respect to the plane defined by all the heavy atoms. The Δz , $\Delta \cos \beta'_A$, and $\Delta \sin \theta$ values are all rms quantities

and are defined as follows: $\Delta z = [\langle z^2 \rangle - \langle z \rangle^2]^{1/2}$, $\Delta \cos \beta'_A = [\langle \cos^2 \beta'_A \rangle - \langle \cos \beta'_A \rangle^2]^{1/2}$, and $\Delta \sin \theta = [\langle \sin^2 \theta \rangle - \langle \sin \theta \rangle^2]^{1/2}$.

The intermolecular vibrational states in Tables V and VI are assigned to the fundamentals, overtones, and combinations of the following five intermolecular modes: (1) the vdW bend mode, ν_b , (2) the water inversion mode, ν_{inv} (also referred to as the water wag), (3) the internal rotation mode, ν_{ir} , (4) the vdW stretch mode, ν_s , and (5) the water rock mode, ν_{rock} . These assignments are made in two ways. The main one is by inspecting the plots of the reduced probability densities (RPDs) of the eigenstates in 1D and 2D along suitably chosen coordinates, as in Figs. 1–6 for the intermolecular fundamentals and one combination state of the D₂O–CO₂ complex. The complementary way of making the assignments is by observing in Tables V and VI how $\langle d \rangle$, Δd , Δz , $\Delta \cos \beta'_A$, and $\Delta \sin \theta$ vary from one state to another. These quantities tend to be rather sensitive indicators of the excitation, or lack thereof, of certain intermolecular modes, as illustrated below.

All the states of D₂O–CO₂ listed in Table V are assigned rather unambiguously. They are discussed in the following. Figure 1 shows the 2D ($z, \cos \beta'_A$) RPD contour plots of four states of the vdW bend (ν_b) mode progression for $\nu_b = 1$ –4. The nodal patterns are clear, making it easy to count the number of quanta in the bend mode. The assignments are corroborated by the changes in $\Delta \cos \beta'_A$,

TABLE VI. Low-energy intermolecular vibrational states of $\text{H}_2\text{O}-\text{CO}_2$ from quantum 5D rigid-monomer calculations. $\langle d \rangle$, Δd , and Δz (all in Å), as well as $\Delta \cos \beta'_A$ and $\Delta \sin \theta$, are defined in the text.

	ΔE^a (δ) ^b	$\langle d \rangle$ (Δd)	Δz	$\Delta \cos \beta'_A$	$\Delta \sin \theta$	Irrep ^c	Assignment
1	0.00 (−0.0723)	2.915 (0.115)	0.268	0.353	0.437	A'_1	Ground state
2	47.59 (0.0488)	2.938 (0.117)	0.442	0.512	0.430	B''_2	ν_b
3	91.20 (−0.0997)	2.976 (0.168)	0.494	0.509	0.435	A'_1	$2\nu_b/\nu_s$
4	103.48 (−0.0705)	2.934 (0.117)	0.275	0.362	0.666	B''_1	ν_{inv}
5	105.31 (−0.1260)	2.975 (0.163)	0.409	0.469	0.434	A'_1	$2\nu_b/\nu_s$
6	116.58 (1.7325)	2.947 (0.118)	0.278	0.438	0.432	A'_2	ν_{ir}
7	131.03 (0.0707)	3.005 (0.182)	0.620	0.591	0.432	B''_2	$3\nu_b$
8	157.69 (0.2897)	2.963 (0.120)	0.450	0.427	0.647	A'_2	$\nu_{\text{inv}} + \nu_b$
9	159.94 (0.1426)	2.977 (0.131)	0.476	0.478	0.432	B''_2	$\nu_s + \nu_b$
10	162.10 (0.2252)	2.985 (0.152)	0.442	0.532	0.432	B''_2	ν_{rock} (tentative)
11	167.56 (−1.2463)	2.979 (0.127)	0.449	0.556	0.451	B''_1	$\nu_{\text{ir}} + \nu_b$
12	169.64 (−0.1137)	3.046 (0.206)	0.735	0.637	0.433	A'_1	$4\nu_b$
13	190.07 (−0.8366)	3.030 (0.217)	0.366	0.466	0.415	A'_1	$2\nu_s$
14	193.60 (0.0000)	3.013 (0.178)	0.360	0.485	0.394	A'_1	uncertain
15	196.12 (−0.1099)	3.001 (0.192)	0.396	0.396	0.654	B''_1	$\nu_{\text{inv}} + \nu_s$
16	204.71 (0.0703)	3.086 (0.229)	0.837	0.675	0.434	B''_2	$5\nu_b$
17	206.28 (2.9413)	3.016 (0.197)	0.409	0.498	0.455	A'_2	$\nu_{\text{ir}} + \nu_s$

^aEnergy in cm^{-1} relative to the ground-state energy of the complex.^bTunneling shift in cm^{-1} .^cIrreducible representation of the group G_8 . See, for example, Ref. 3.

which increases steadily for the members of the bend-mode progression from 0.501 for $\nu_b = 1$, the bend-mode fundamental at 36.09 cm^{-1} , to 0.704 for $\nu_b = 5$ at 181.09 cm^{-1} . In contrast, $\Delta \cos \beta'_A$ is much smaller for the states that do not have a contribution from the bend mode. It is easy to see from Table V that the energy differences between the neighboring bend states decrease slowly with the increasing number of quanta. Thus, the vdW bend exhibits weak positive anharmonicity.

Figure 2 shows the 2D (ρ, z) RPD contour plots of the vdW stretch states ν_s and $2\nu_s$ at 97.47 and 190.84 cm^{-1} , respectively; the stretch mode shows weak positive anharmonicity as well. The coordinate ρ on the abscissa of Fig. 2 is a cylindrical coordinate defined in Eq. (16); it is perpendicular to z and essentially represents the vdW stretch coordinate of the complex. The nodal patterns of both states are regular, making their assignment straightforward.

Figure 3 shows the 1D RPDs of the internal-rotation states ν_{ir} and $2\nu_{\text{ir}}$ at 86.80 and 162.09 cm^{-1} , respectively; this mode also exhibits weak positive anharmonicity. Shown as well for comparison are the 1D RPDs of the ground state and ν_b . The γ'_A Euler angle on the abscissa of the left column refers to the rotation about the water symmetry axis in the CO_2 -embedded dimer-fixed frame of Sec. II C. On the abscissa of the right column, γ_A has the same role but in the Brocks-type approach of Sec. II B. The comparison of the RPDs in the two columns makes it clear that the new approach, in the left column, is superior in identifying the internal-rotation states, and the number of quanta in them, and distinguishing them from other intermolecular modes. On the other hand, the RPDs from the Brocks-type approach in the right column are qualitatively the same for the ground state and the two internal-rotation states and would be unable to flag the latter. The implication is that in the new

approach, the internal-rotation coordinate is described well by γ'_A alone, while in the Brocks-type approach, it is a mix of γ_A and other coordinates.

The 2D RPD contour plot in ($z, \cos \beta'_A$) of the water rock-mode fundamental ν_{rock} , at 140.99 cm^{-1} , is displayed in Fig. 4. It is accompanied by the contour plot of the vdW bend fundamental (ν_b). The comparison of the two contour plots reveals that (a) ν_{rock} includes appreciable water bend motion corresponding to the water symmetry axis rotating relative to the CO_2 internuclear axis, (b) this motion is accompanied by less motion of the water c.m. along the CO_2 axis in comparison to ν_b , and (c) to the extent that there is translational motion of the water c.m. along the CO_2 axis, it is out of phase with the water bending motion (relative to ν_b).

Figure 5 shows the 1D RPDs of the water inversion states ν_{inv} and $2\nu_{\text{inv}}$ at 72.95 and 156.54 cm^{-1} , respectively, together with that of the ground states. These highly regular plots as a function of the out-of-plane angle θ clearly identify the inversion mode and the number of quanta in it. This means that θ provides an excellent description of the water inversion motion. Moreover, for the inversion states in Table V, $\Delta \sin \theta$ has much larger values than for the states with little or no component of the inversion mode, making them readily recognizable. The energy difference between ν_{inv} and $2\nu_{\text{inv}}$, 83.56 cm^{-1} , is greater than the energy of ν_{inv} , meaning that this mode exhibits negative anharmonicity, in contrast to ν_b , ν_s , and ν_{ir} modes. Negative anharmonicity was also observed for the water inversion mode of the $\text{HCl}-\text{H}_2\text{O}$ complex.¹⁴

Figure 6 shows both the 2D ($z, \cos \beta'_A$) contour plot and the 1D RPD along θ for the state $2\nu_{\text{inv}} + \nu_b$ at 201.55 cm^{-1} . It illustrates the utility of such plots in the assignment of the combination states involving the excitation of two different modes. The 2D contour plot

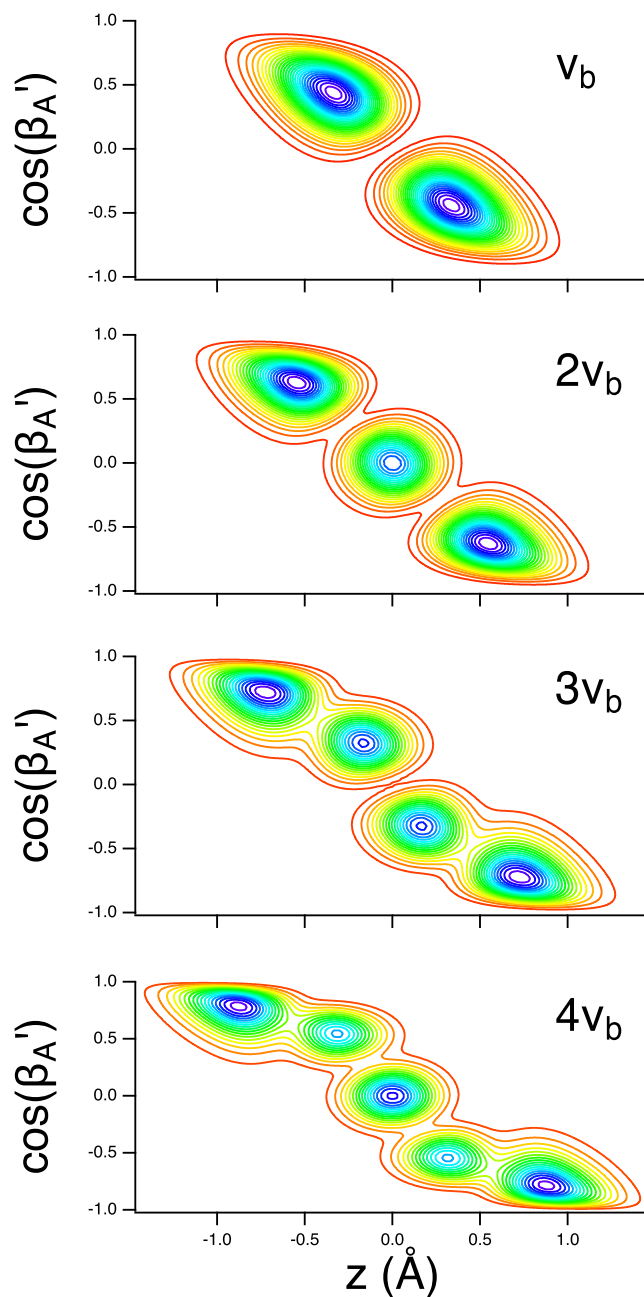


FIG. 1. Contour plots of the reduced probability densities of the D₂O–CO₂ vdW bend (ν_b) progression with one to four quanta as a function of the coordinates $\cos \beta_A'$ and z defined in the text.

at the top of the figure unmistakably resembles that of ν_b in Fig. 1, while the 1D RPD at the bottom of this figure bears uncanny resemblance to that for $2\nu_{\text{inv}}$ in Fig. 5. Other combination states can be assigned in the same manner by inspecting 1D and 2D plots of their RPDs as a function of the appropriately chosen coordinates.

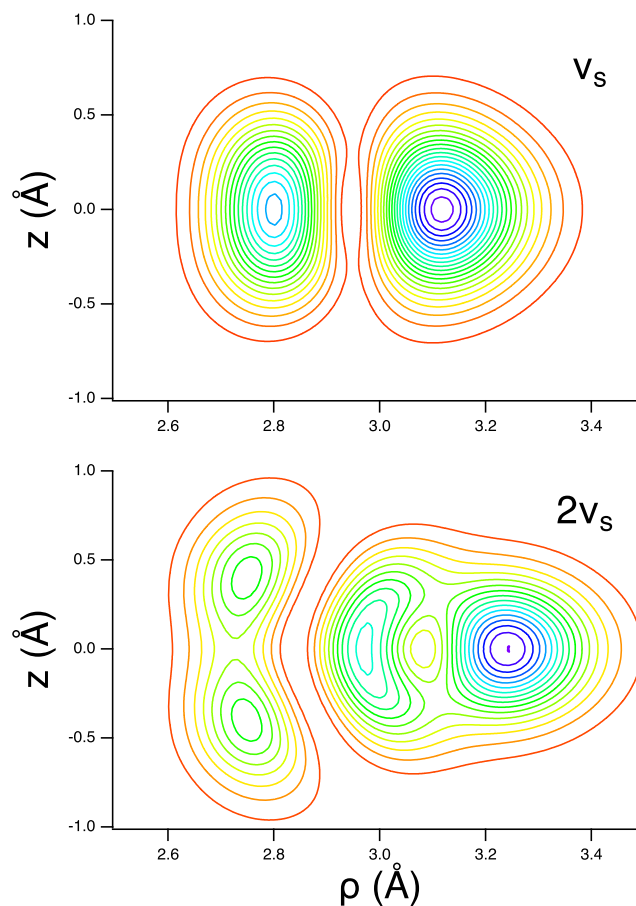


FIG. 2. Contour plots of the reduced probability densities of the D₂O–CO₂ vdW stretch (ν_s) states with one and two quanta as a function of the coordinates z and ρ defined in the text.

The intermolecular vibrational states of the H₂O–CO₂ complex in Table VI share most of the characteristics of the D₂O–CO₂ eigenstates discussed above. Therefore, there is no need to elaborate on them again. However, the eigenstates of H₂O–CO₂ do exhibit one distinctive, and complicating, feature not present in D₂O–CO₂. It is the Fermi resonance between the bend overtone ($2\nu_b$) and the stretch fundamental (ν_s). This gives rise to the pair of states designated $2\nu_b/\nu_s$ at 91.20 and 105.31 cm^{−1}, respectively. The $2\nu_b/\nu_s$ coupling precludes unambiguous determination of the ν_s fundamental for H₂O–CO₂. It also introduces uncertainties in the assignments of the combination states involving the vdW stretch and/or bend modes.

In Sec. III A, we discussed the rotation-tunneling states of H₂O–CO₂ and D₂O–CO₂ complexes in their ground vibrational states, listed in Tables I and II, respectively. From the tunneling shifts δ given for each intermolecular vibrational state in Tables V and VI, one can see how the corresponding tunneling splittings, equal to $2|\delta|$, are affected by the excitation of different vibrational modes of the complexes.

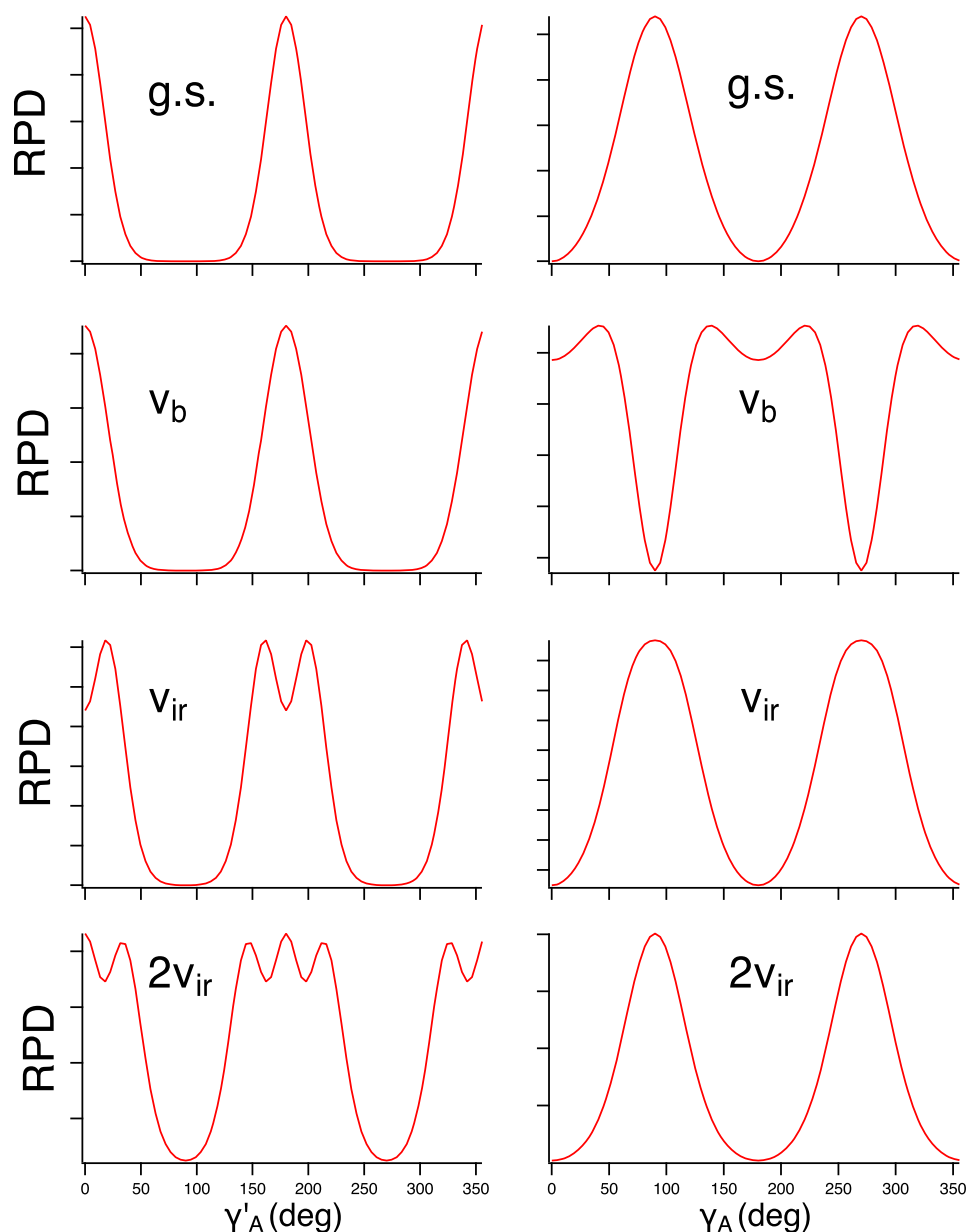


FIG. 3. Reduced probability densities (RPDs) of the $\text{D}_2\text{O}-\text{CO}_2$ ground state (g.s.), the vdW bend (v_b) fundamental, and the internal-rotation (v_{ir}) states with one and two quanta. In the left column, the RPDs are shown as a function of the coordinate γ'_A and in the right column as a function of the coordinate γ_A ; both are defined in the text.

Excitation of the internal-rotation mode greatly increases the magnitude of δ , and the tunneling splitting, in both isotopologues. Thus, for $\text{D}_2\text{O}-\text{CO}_2$, $v_{ir}|\delta|$ equals 0.0633 cm^{-1} , increasing dramatically to 0.8538 cm^{-1} for $2v_{ir}$. Both values are far larger than $|\delta| = 0.0019\text{ cm}^{-1}$ in the ground state and those for other excited intermolecular modes. The same holds for the $\text{H}_2\text{O}-\text{CO}_2$ complex. For v_{ir} , $|\delta| = 1.7325\text{ cm}^{-1}$, compared to 0.0723 cm^{-1} in the ground state. Of course, it is not surprising that the excitation of the

internal-rotation mode that gives rise to the tunneling splitting increases the magnitude of the splitting.

Inspection of [Tables V](#) and [VI](#) reveals that the tunneling splittings are indeed highly mode-specific, i.e., depend much more on the nature of the given state than on its energy. For example, in [Table V](#), $|\delta|$ of v_{ir} at 86.80 cm^{-1} (0.0633 cm^{-1}) is much larger than that of v_s (0.0095 cm^{-1}), although the latter is higher energy, at 97.47 cm^{-1} . Likewise, in [Table VI](#), $|\delta|$ of v_{ir} at 116.58 cm^{-1} (1.7325 cm^{-1}),

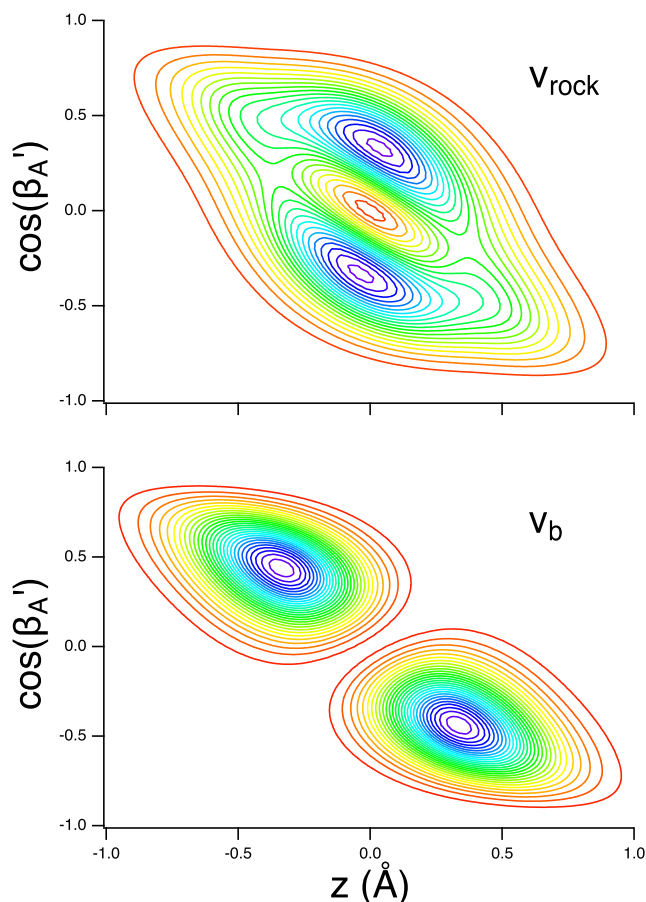


FIG. 4. Top: contour plot of the D₂O–CO₂ rock (ν_{rock}) fundamental as a function of the coordinates $\cos \beta'_A$ and z defined in the text. Bottom: contour plot of the vdW bend (ν_b) fundamental also as a function of the coordinates $\cos \beta'_A$ and z defined in the text.

is much larger than that of $3\nu_b$ (0.0707 cm^{-1}), at the higher energy of 131.03 cm^{-1} .

It is interesting that the H₂O–CO₂ combination state $\nu_{\text{ir}} + \nu_s$ has the tunneling shift of 2.9413 cm^{-1} , which is significantly larger than that for ν_{ir} alone (1.7325 cm^{-1}). The same is observed in D₂O–CO₂. The tunneling shift of its $\nu_{\text{ir}} + \nu_s$ state (0.2591 cm^{-1}) is much larger than that for ν_{ir} (0.0633 cm^{-1}). A tentative explanation for this is that exciting the vdW stretch mode ν_s , simultaneously with the ν_{ir} excitation, increases the average distance between the CO₂ and water moieties in the two complexes, thereby decreasing somewhat the barrier to the internal rotation.

Table VII summarizes the calculated energies of the key intermolecular vibrational states of H₂O–CO₂ and D₂O–CO₂ complexes. These include the fundamentals of all five intermolecular vibrational modes of D₂O–CO₂ and the fundamentals of four out of five intermolecular vibrational modes of H₂O–CO₂; its ν_s fundamental is uncertain owing to the $2\nu_b/\nu_s$ Fermi resonance. Included also are the $2\nu_b$, $2\nu_s$, and $2\nu_{\text{inv}}$ overtones of both complexes and $2\nu_{\text{ir}}$ for

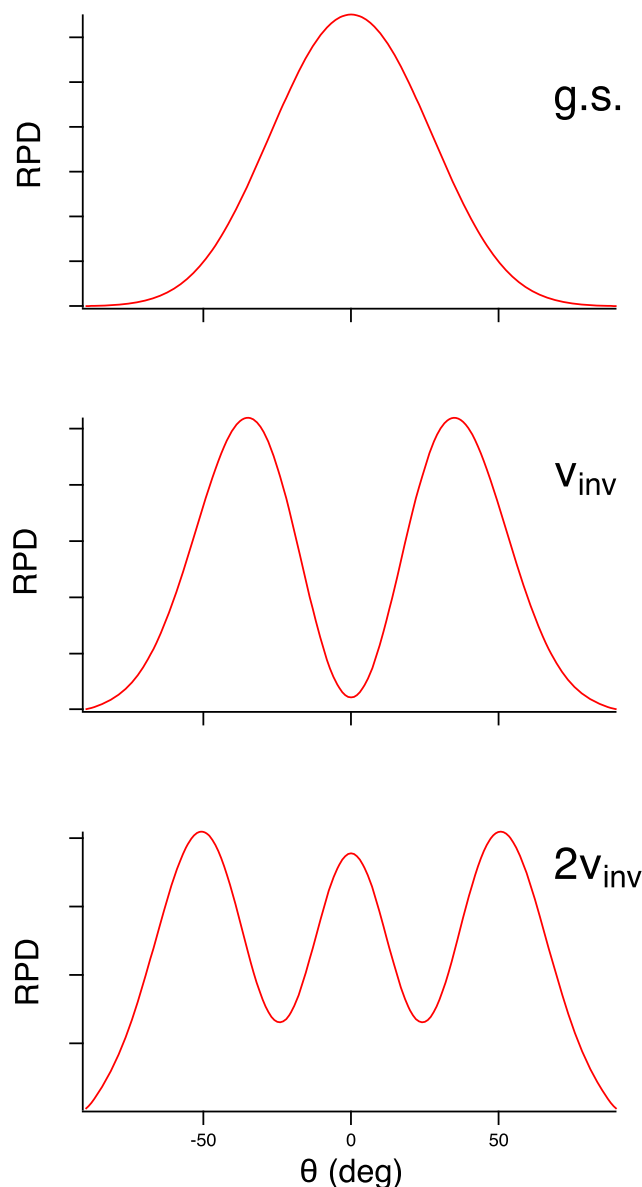


FIG. 5. Reduced probability densities (RPDs) of the D₂O–CO₂ ground state (g.s.) and the inversion (ν_{inv}) states with one and two quanta as a function of the coordinate θ defined in the text.

D₂O–CO₂ only. The only experimental results available for comparison are those measured for these complexes in neon matrices at 2.8 K;⁷ they are included in Table VII. For ν_{inv} of H₂O–CO₂ and ν_{rock} of both H₂O–CO₂ and D₂O–CO₂, our calculations and experiment agree to within $2\text{--}4\text{ cm}^{-1}$. This level of agreement is good, given that the rigid-monomer approximation on the theory side and the neon-matrix shifts in the experiments are likely to introduce uncertainties of this magnitude in the comparison.

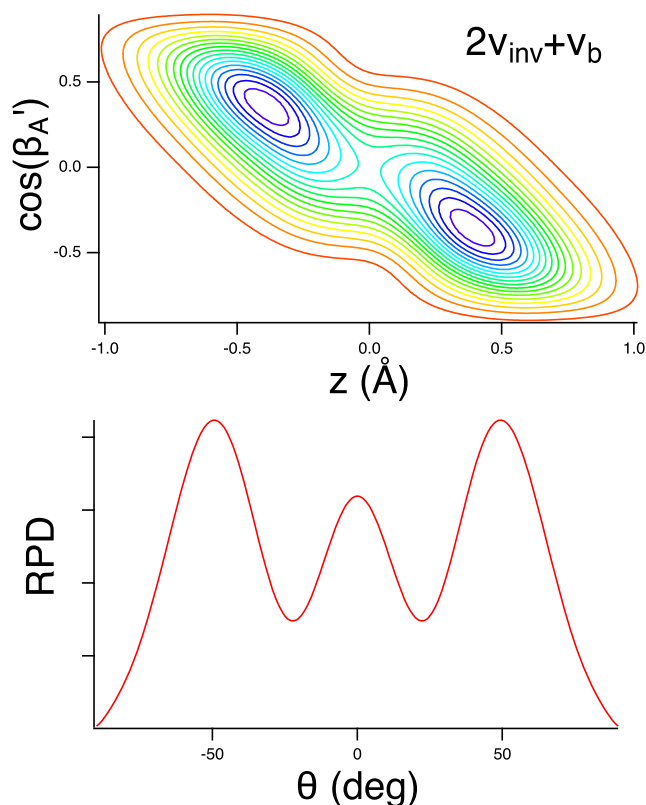


FIG. 6. The $\text{D}_2\text{O}-\text{CO}_2$ state $2\nu_{\text{inv}} + \nu_b$: Top: contour plot as a function of the coordinates $\cos \beta_A'$ and z defined in the text. Bottom: reduced probability density (RPD) as a function of the coordinate θ defined in the text.

TABLE VII. Summary of the calculated energies of key intermolecular vibrational states (in cm^{-1}) of $\text{D}_2\text{O}-\text{CO}_2$ and $\text{H}_2\text{O}-\text{CO}_2$ complexes from Tables V and VI, respectively. The numbers in parentheses are the corresponding experimental values measured for these complexes in neon matrices at 2.8 K from Ref. 7.

	$\text{H}_2\text{O}-\text{CO}_2$	$\text{D}_2\text{O}-\text{CO}_2$
ν_b	47.59	36.09
$2\nu_b$	91.20/105.31	75.81
ν_s	91.20/105.31	97.47
$2\nu_s$	190.07	190.84
ν_{inv}	103.48 (101.4)	72.95
$2\nu_{\text{inv}}$	223.84	156.54
ν_{ir}	116.58	86.80
$2\nu_{\text{ir}}$		162.09
ν_{rock}	162.10 (164.7)	140.99 (145.1)

IV. CONCLUSIONS

In this work, we have presented the quantum 5D calculations of the fully coupled intermolecular rovibrational states of the important $\text{H}_2\text{O}-\text{CO}_2$ and $\text{D}_2\text{O}-\text{CO}_2$ vdW complexes in the rigid-monomer approximation for the total angular momentum

J values of 0, 1, and 2. The rigid-monomer version of the full-dimensional (12D) PES by Wang and Bowman¹¹ is utilized. These rigorous bound-state calculations yield detailed information, lacking until now, regarding the rovibrational eigenstates of the two isotopologues and their assignments, the internal-rotation tunneling splittings, and how they vary with both rotational and vibrational excitations, as well as the rotational constants of the two complexes.

The calculations for the two heavy (CO_2)-light ($\text{H}_2\text{O}/\text{D}_2\text{O}$) weakly bound complexes are performed using two approaches, which differ in the definition of the dimer-fixed (DF) frame, and the associated coordinates, employed. In the first, introduced by Brocks *et al.*²⁰ and utilized in all our recent quantum calculations of the (ro)vibrational states of noncovalently bound triatom-diatom complexes,^{12–16} the z axis of the DF frame is oriented along the vector pointing from the c.m. of one of the monomers to that of the other. In the second approach introduced in this paper, we derive and implement the new rovibrational Hamiltonian based on the DF frame embedded in the CO_2 (heavy) moiety so that its z axis is parallel to the internuclear axis of the CO_2 moiety. We demonstrate that this novel approach gives results whose accuracy matches that from the Brocks-type approach for basis sets whose sizes are less than a half of the basis-set sizes employed in the latter approach. Moreover, it turns out that the intermolecular eigenfunctions computed employing the CO_2 -embedded DF frame in many instances exhibit nodal patterns that are more regular and informative than those obtained in the Brocks coordinate system, allowing them to be interpreted and assigned with confidence. We believe that this approach should be preferred when dealing with weakly bound molecular dimers comprised of a heavy and a light monomer.

In the vibrational ground state of $\text{H}_2\text{O}-\text{CO}_2$, the internal-rotation tunneling splitting calculated for the lowest rotational state 0_{00} of the complex, 0.1447 cm^{-1} , agrees very well, within 6%, with the experimentally determined tunneling splitting of $0.1356(16) \text{ cm}^{-1}$.² As expected, the corresponding splitting calculated for $\text{D}_2\text{O}-\text{CO}_2$ is much smaller, 0.0039 cm^{-1} . Also pertaining to the vibrational ground state, a number of rotational transition frequencies and the three rotational constants calculated for $\text{H}_2\text{O}-\text{CO}_2$ and $\text{D}_2\text{O}-\text{CO}_2$, respectively, are in excellent agreement with the corresponding experimental values.^{1,3}

For excitation energies up to about 200 cm^{-1} above the ground state, the computed intermolecular vibrational states of $\text{D}_2\text{O}-\text{CO}_2$ and $\text{H}_2\text{O}-\text{CO}_2$ complexes are assigned in terms of the fundamentals, overtones, and combinations of their five intermolecular modes: the vdW bend, the vdW stretch, the internal-rotation mode, the water rock, and the water inversion. In addition, for each of these states, their tunneling shifts δ and the associated tunneling splittings equal to $2|\delta|$ are calculated. They are mode-specific, i.e., dependent much more on the character of the excited states than on their energies. In particular, excitation of the internal-rotation mode greatly increases the magnitude of the tunneling shift/splitting relative to that of the ground state and those of other excited vibrational modes.

Our calculations yield the fundamentals of all five intermolecular vibrational modes of the $\text{D}_2\text{O}-\text{CO}_2$ complex. In the case of the $\text{H}_2\text{O}-\text{CO}_2$ complex, four out of five intermolecular vibrational fundamentals are characterized; the energy of the vdW stretch fundamental cannot be determined unambiguously due to its Fermi resonance with the vdW bend overtone. The experimental results available for direct comparison are scant and come only from the

terahertz absorption spectra of these complexes in cryogenic neon matrices.⁷ They agree with our results to within $2\text{--}4\text{ cm}^{-1}$, which is very satisfactory in view of the rigid-monomer approximation made in the calculations and the matrix-induced shifts present in the spectra.

It is our hope that the comprehensive theoretical characterization of the intermolecular rovibrational states of $\text{H}_2\text{O}\text{--}\text{CO}_2$ and $\text{D}_2\text{O}\text{--}\text{CO}_2$ in this paper, including their internal-rotation tunneling splittings, will motivate new spectroscopic investigations of the intermolecular vibrations of these vdW complexes in the gas phase, free from matrix shifts. They would allow a more conclusive and detailed assessment of the accuracy of the PES employed and, if necessary, guide its improvement.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

DATA AVAILABILITY

The data that support the findings of this study are available within the article.

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